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DETERMINATION OF A SAFETY ISSUE RELATING TO A VEHICLE OCCUPANT

Abstract

Systems, methods, and other embodiments described herein relate to determining a safety issue relating to an occupant of a vehicle. In one embodiment, a method includes requesting, in response to a portion of the occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin. The method further includes determining, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.

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Background/Summary

TECHNICAL FIELD

[0001] The subject matter described herein relates, in general, to systems and methods for determining whether there is a safety issue relating to an occupant of a vehicle.

BACKGROUND

[0002] Modern vehicles include seatbelt sensors for detecting whether a seatbelt is securely fastened. The seatbelt sensors may determine that the seatbelt is securely fastened based on the latch plate being locked into the buckle. However, the seatbelt sensors may not accurately detect whether the occupant has the seatbelt straps properly placed across their torso.

SUMMARY

[0003] This section generally summarizes the disclosure and is not a comprehensive explanation of its full scope or all its features.

[0004] In another embodiment, a system for determining whether there is a safety issue relating to an occupant of a vehicle is disclosed. The system includes a processor and a memory in communication with the processor. The memory stores machine-readable instructions that, when executed by the processor, cause the processor to request, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin, and determine, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.

[0005] In one embodiment, a method for determining whether there is a safety issue relating to an occupant of a vehicle is disclosed. The method includes requesting, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin, and determining, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.

[0006] In another embodiment, a non-transitory computer-readable medium for determining whether there is a safety issue relating to an occupant of a vehicle and including instructions that, when executed by a processor, cause the processor to perform one or more functions, is disclosed. The instructions include instructions to request, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin, and determine, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

[0008] FIG. 1 illustrates a block diagram of a vehicle incorporating a vehicle occupant safety

determination system.

[0009] FIGS. 2A-2C illustrate an example of a vehicle incorporating a vehicle occupant safety determination system as well as the first and second sensor data.

[0010] FIG. 3 is a more detailed block diagram of the vehicle occupant safety determination system of FIG. 2.

[0011] FIG. 4 is a flowchart illustrating one embodiment of a method associated with determining whether there is a safety issue relating to an occupant of a vehicle.

DETAILED DESCRIPTION

[0012] Systems, methods, and other embodiments associated with determining a safety issue relating to one or more occupants of a vehicle are disclosed. Occupant(s) in a vehicle may not have their seatbelts securely fastened or even if their seatbelts are fastened, occupant(s) may not have their seatbelts positioned properly across their body. In some cases, occupant(s) may not be sitting safely in their seats. As an example, a child passenger may have unfastened their seatbelt or moved the seatbelt strap from their torso. All these instances may cause a safety issue and may pose a risk to the occupant(s).

[0013] A current method for detecting whether a seatbelt of the occupant is securely fastened includes the use of seatbelt sensors to determine whether the latch plate of the seatbelt is locked into the buckle of the seatbelt. However, an occupant may fasten the seatbelt and place a portion of the seatbelt strap, such as the portion that rests across the torso, behind their back. In such a case, the seatbelt sensor is unable to detect that the seat belt of the occupant is not securely fastened across the occupant.

[0014] Another current method for detecting whether the seatbelt of the occupant is securely fastened includes using vehicle cameras. However, in a case where the occupant is holding an object such as a laptop or a large book, the images or videos captured by the vehicle cameras may not include the torso of the occupant which is occluded by the object, which is between the vehicle camera and the occupant. Another current method for detecting whether the seatbelt of the occupant is securely fastened includes using radar sensors. However, in a case where the occupant is holding an object that has metal such as a laptop, the radar sensor is unable to pass through metal and as such, a portion of the occupant would be occluded by the metal in the object. These current methods may be unable to detect and determine whether the occupant has their seatbelt safely and securely fastened.

[0015] Accordingly, in one embodiment, the disclosed approach is a system that determines whether there is a safety issue relating to one or more occupants of a vehicle. The system may include one or more vehicle sensors capable of capturing first sensor data relating to the occupant(s) in the vehicle, object(s) in the vehicle, and/or the vehicle cabin. Examples of vehicle sensors include radar sensors, LiDAR sensors, thermal sensors, and/or cameras. The system requests and receives first sensor data from the vehicle sensors. The system determines whether the seatbelts of the occupants are securely fastened and are properly placed across the body of the occupant. The system may further determine whether the occupants are sitting properly in their seats. The system may determine that there are one or more occupants for which the system cannot determine the positioning or placement of the seatbelt and/or cannot determine whether the occupant is sitting down properly because an object occludes a portion of the occupant. As an example, the object may be a mobile device such as a mobile phone, a computer, a laptop, a tablet, a camera, a display unit, and/or a video player. More generally, the object may be anything that the vehicle sensor(s) cannot perceive through.

[0016] The system may then detect one or more mobile devices that are capable of capturing second sensor data in the vehicle. The second sensor data also relates to the occupant(s) in the vehicle, object(s) in the vehicle, and/or the vehicle cabin. However, based on the location of the mobile device, the sensors in the mobile device may be capable of capturing second sensor data that differs from the first sensor data in terms of perspective and/or field of view. The system may

communicate with one or more of the mobile devices, requesting the second sensor data. In one case, the system may determine which mobile device to communicate with based on the location of the mobile device within the vehicle cabin. In another case, the system may communicate with the mobile devices in a round-robin manner, determining which of the mobile devices can provide second sensor data in which the portion of the occupant that is occluded in the first sensor data is visible.

[0017] The system may then determine whether there is a safety issue in which the seatbelts of the occupants are properly placed, securely fastened and/or the occupants are properly seated. The system may issue a notification indicating that there is or there is no safety issue based on the system determining that there is or there is no safety issue. Additionally, the system may operate the vehicle based upon the determination that there is or there is no safety issue. As an example, the system may prevent the vehicle from being started until the safety issues have been resolved. The system may be utilized when the vehicle is stationary or when the vehicle is in motion. The system may periodically determine whether there is a safety issue multiple times while the vehicle is in motion. Further, the system may be utilized for determining a safety issue pertaining to one or more occupants at a time.

[0018] The embodiments disclosed herein present various advantages over the current methods. First, the embodiments determine whether the seatbelt of one or more occupants is properly fastened and positioned on the occupant with higher accuracy than the current methods. Second, the embodiments control the vehicle based on whether there is a safety issue or there is no safety issue. Third, the embodiments utilize sensors in mobile devices within the vehicle and do not require additional hardware or computing resources.

[0019] Detailed embodiments are disclosed herein; however, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in the figures, but the embodiments are not limited to the illustrated structure or application.

[0020] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details.

[0021] Referring to FIG. 1, a block diagram of the vehicle **102** incorporating a vehicle occupant safety determination system **100** is illustrated. The vehicle **102** includes various elements. It will be understood that in various embodiments, it may not be necessary for the vehicle **102** to have all of the elements shown in FIG. 1. The vehicle **102** can have any combination of the various elements shown in FIG. 1. Further, the vehicle **102** can have additional elements to those shown in FIG. 1. In some arrangements, the vehicle **102** may be implemented without one or more of the elements shown in FIG. 1. While the various elements are shown as being located within the vehicle **102** in FIG. 1, it will be understood that one or more of these elements can be located external to the vehicle **102**. Further, the elements shown may be physically separated by large distances. For example, one or more components of the vehicle occupant safety determination system **100** can be implemented within the vehicle **102** while further components of the vehicle occupant safety determination system **100** are implemented within a cloud-computing environment. In such an example, the vehicle occupant safety determination system **100** may acquire data from a mobile device and the vehicle **102** and execute as a cloud-based resource that is comprised of devices (e.g.,

distributed servers) remote from the vehicle **102** to determine a safety issue relating to one or more occupants of the vehicle **102**. It should be appreciated that apportionment of the processing between the vehicle **102** and a cloud server may vary according to different implementations. [0022] In one or more embodiments, the vehicle **102** can be a conventional vehicle that is configured to operate in only a manual mode. In one or more embodiments, the vehicle **102** can be an autonomous vehicle. As used herein, “autonomous vehicle” refers to a vehicle that operates in an autonomous mode. “Autonomous mode” refers to navigating and/or maneuvering the vehicle **102** along a travel route using one or more computing systems to control the vehicle **102** with minimal or no input from a human driver. In one or more embodiments, the vehicle **102** can be highly automated or completely automated. In one or more embodiments, the vehicle **102** can be configured with one or more semi-autonomous operational modes in which one or more computing systems perform a portion of the navigation and/or maneuvering of the vehicle **102** along a travel route, and a vehicle operator (i.e., driver) provides inputs to the vehicle **102** to perform a portion of the navigation and/or maneuvering of the vehicle **102** along a travel route.

[0023] The vehicle **102** can include one or more processors **110**. In one or more arrangements, the processor(s) **110** can be a main processor of the vehicle **102**. For instance, the processor(s) **110** can be an electronic control unit (ECU). The vehicle **102** can include one or more data stores **115** for storing one or more types of data. The data store **115** can include volatile and/or non-volatile memory. Examples of suitable data stores **115** include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. The data store **115** can be a component of the processor(s) **110**, or the data store **115** can be operatively connected to the processor(s) **110** for use thereby. The term “operatively connected,” as used throughout this description, can include direct or indirect connections, including connections without direct physical contact.

[0024] In one or more arrangements, the data store(s) **115** can include sensor data **130**. The sensor data **130** may include first sensor data **132** that originates from a sensor system **120** of the vehicle **102** and second sensor data **134** that originates from one or more mobile devices. The first sensor data **132** may include sensor data from vehicle sensors **121**, environment sensors **122**, and/or any other suitable sensors in the vehicle **102**. The second sensor data **134** may include sensor data from mobile devices such as a mobile phone, a computer, a laptop computer, a tablet computer, a display unit, a camera, and/or a video player such as a DVD player. In general, the mobile device may include any device capable of capturing an image and/or a video. In some instances, one portion of the sensor data **130** may be located in the data store(s) **115** located onboard the vehicle **102** and another portion of the sensor data **130** may be located in storage outside the vehicle **102**.

[0025] The vehicle **102** may include the sensor system **120**. The sensor system **120** can include one or more sensors. “Sensor” means any device, component and/or system that can detect, and/or sense something. The one or more sensors can be configured to detect, and/or sense in real-time. As used herein, the term “real-time” means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor **110** to keep up with some external process.

[0026] In arrangements in which the sensor system **120** includes a plurality of sensors, the sensors can work independently from each other. Alternatively, two or more of the sensors can work in combination with each other. In such a case, the two or more sensors can form a sensor network. The sensor system **120** and/or the one or more sensors can be operatively connected to the processor(s) **110**, the data store(s) **115**, and/or another element of the vehicle **102** (including any of the elements shown in FIG. 1).

[0027] The sensor system **120** can include any suitable type of sensor. Various examples of different types of sensors will be described herein. However, it will be understood that the

embodiments are not limited to the particular sensors described. The sensor system **120** can include one or more vehicle sensors **121**. The vehicle sensor(s) **121** can detect, determine, and/or sense information about the vehicle **102** and/or an event inside the vehicle **102**, such as in the vehicle cabin. The vehicle sensor(s) **121** can detect and/or record the time an event occurred. In one or more arrangements, the vehicle sensor(s) **121** can be configured to detect and/or sense position, orientation, and movement of occupants of the vehicle **102**.

[0028] The vehicle sensor(s) **121** can include one or more radar sensors **123**. The radar sensor(s) **123** can be located in the vehicle cabin and can detect the position and/or the movement of occupant(s) inside the vehicle cabin. The vehicle sensor(s) **121** can include one or more LiDAR sensors **124**. The LiDAR sensor(s) **124** can be located in the vehicle cabin and can detect the position and/or the movement of occupant(s) inside the vehicle cabin. The vehicle sensor(s) **121** can include one or more thermal sensors **125**. The thermal sensor(s) **125** can be located in the vehicle cabin and can detect the position and/or the movement of occupant(s) inside the vehicle cabin based on temperature.

[0029] The vehicle sensor(s) **121** can include one or more cameras **126**. The camera(s) **126** can capture the position and/or the movement of the occupant(s) inside the vehicle cabin. The cameras **126** may capture images and/or video. In one or more arrangements and as example, the cameras **126** may be high dynamic range (HDR) cameras or infrared (IR) cameras.

[0030] Alternatively, or in addition, the sensor system **120** can include one or more environment sensors **122** configured to acquire, and/or sense driving environment data. “Driving environment data” includes data or information about the external environment in which the vehicle is located or one or more portions thereof. For example, the one or more environment sensors **122** can be configured to detect, measure, quantify and/or sense other objects in the external environment of the vehicle **102**.

[0031] The vehicle **102** can include one or more vehicle systems **140**. Various examples of the one or more vehicle systems **140** are shown in FIG. 1. However, the vehicle **102** can include more, fewer, or different vehicle systems **140**. It should be appreciated that although particular vehicle systems are separately defined, each or any of the systems or portions thereof may be otherwise combined or segregated via hardware and/or software within the vehicle **102**. The vehicle **102** can include a propulsion system **141**, a braking system **142**, a steering system **143**, throttle system **144**, a transmission system **145**, a signaling system **146**, and/or a navigation system **147**. Each of these systems can include one or more devices, components, and/or a combination thereof, now known or later developed.

[0032] The vehicle **102** can include one or more actuators **150**. The actuators **150** can be any element or combination of elements operable to modify, adjust and/or alter one or more of the vehicle systems **140** or components in response to receiving signals or other inputs from the processor(s) **110**, the vehicle occupant safety determination system **100**, and/or the autonomous driving module(s) **160**. Any suitable actuator can be used. For instance, the one or more actuators **150** can include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and/or piezoelectric actuators, just to name a few possibilities.

[0033] The vehicle **102** can include an input system **133**. An “input system” includes any device, component, system, element or arrangement or groups thereof that enable information/data to be entered into a machine. The input system **133** can receive an input from a user (e.g., a driver or a passenger). The vehicle **102** can include an output system **135**. An “output system” includes any device, component, or arrangement or groups thereof that enable information/data to be presented to a user (e.g., a person, a vehicle passenger, etc.) such as a display interface.

[0034] The vehicle **102** can include one or more modules, at least some of which are described herein. The modules can be implemented as computer-readable program code that, when executed by a processor **110**, implement one or more of the various processes described herein. One or more of the modules can be a component of the processor(s) **110**, or one or more of the modules can be

executed on and/or distributed among other processing systems to which the processor(s) **110** is operatively connected. The modules can include instructions (e.g., program logic) executable by one or more processor(s) **110**. Alternatively, or in addition, one or more data store **115** may contain such instructions.

[0035] In one or more arrangements, one or more of the modules described herein can include artificial or computational intelligence elements, e.g., neural network, fuzzy logic, or other machine learning algorithms. Further, in one or more arrangements, one or more of the modules can be distributed among a plurality of the modules described herein. In one or more arrangements, two or more of the modules described herein can be combined into a single module.

[0036] The vehicle **102** can include one or more autonomous driving modules **160**. The autonomous driving module(s) **160** either independently or in combination with the vehicle occupant safety determination system **100** can be configured to determine travel path(s), current autonomous driving maneuvers for the vehicle **102**, future autonomous driving maneuvers and/or modifications to current autonomous driving maneuvers based on data acquired by the sensor system **120**, driving scene models, and/or data from any other suitable source such as determinations from the sensor data **130**. “Driving maneuver” means one or more actions that affect the movement of a vehicle. Examples of driving maneuvers include accelerating, decelerating, braking, turning, moving in a lateral direction of the vehicle **102**, changing travel lanes, merging into a travel lane, and/or reversing, just to name a few possibilities. The autonomous driving module(s) **160** can be configured to implement determined driving maneuvers. The autonomous driving module(s) **160** can cause, directly or indirectly, such autonomous driving maneuvers to be implemented. As used herein, “cause” or “causing” means to make, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner. The autonomous driving module(s) **160** can be configured to execute various vehicle functions and/or to transmit data to, receive data from, interact with, and/or control the vehicle **102** or one or more systems thereof (e.g., one or more of vehicle systems **140**).

[0037] As previously mentioned, the vehicle **102** can include the vehicle occupant safety determination system **100**. The processor(s) **110** and/or the vehicle occupant safety determination system **100** can be operatively connected to communicate with the various vehicle systems **140** and/or individual components thereof. The vehicle occupant safety determination system **100** can determine a safety issue relating to one or more occupants in the vehicle **102**. As an example, a safety issue may be fastening of a seat belt, positioning of the seat belt, and/or positioning of the one or more occupants.

[0038] In general, the vehicle occupant safety determination system **100** may utilize first sensor data **132** to determine whether there is a safety issue in relation to one or more occupants. The first sensor data **132** may originate from one or more vehicle sensors **121** and may include a view of one or more occupants. However, a portion of the occupants may be occluded by any number of objects such as a mobile device located between the vehicle sensor(s) **121** and the occupant(s). In a case where a portion of at least one of the occupants is occluded, the vehicle occupant safety determination system **100** may identify a device capable of capturing second sensor data **134** in the vehicle cabin. The vehicle occupant safety determination system **100** may request second sensor data **134** from the device and may then utilize the second sensor data **134** to determine whether there is a safety issue. The device may be the mobile device located between the vehicle sensor(s) **121** and the occupant(s).

[0039] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various

combinations of these elements. In any case, as illustrated in the embodiment of FIG. 1, the vehicle **102** includes a vehicle occupant safety determination system **100** that is implemented to perform methods and other functions as disclosed herein for determining whether there is a safety issue relating to an occupant of a vehicle. As will be discussed in greater detail subsequently, the vehicle occupant safety determination system **100**, in various embodiments, may be implemented partially within the vehicle **102** and may further exchange communications with additional aspects of the vehicle occupant safety determination system **100** that are remote from the vehicle **102** in support of the disclosed functions. Thus, while FIG. 3 generally illustrates the vehicle occupant safety determination system **100** as being self-contained, in various embodiments, the vehicle occupant safety determination system **100** may be implemented within multiple separate devices some of which may be remote from the vehicle **102**.

[0040] Referring to FIGS. 2A-2C, an example of a vehicle **102** incorporating a vehicle occupant safety determination system **100** as well as the first and second sensor data **132**, **134** are illustrated. As previously mentioned, the vehicle **102** may include the vehicle occupant safety determination system **100** and one or more vehicle sensors **121**. As shown in FIG. 2A, at least one or more occupants **202** are inside the vehicle. Additionally, at least one or more mobile devices **204** are inside the vehicle **102**. For simplicity, a single occupant **202** holding a laptop **204** is shown in FIG. 2A.

[0041] As previously explained, the vehicle sensors **121** may include one or more of radar sensors **123**, LiDAR sensors **124**, thermal sensors **125**, and/or cameras **126**. The vehicle sensors **121** are located inside the vehicle cabin **206**. The vehicle sensors **121** are capable of capturing first sensor data **132** relating to the vehicle cabin **206** and the occupant(s) **202** in the vehicle cabin **206**. As such, the vehicle occupant safety determination system **100** may activate one or more of the vehicle sensors **121** to capture first sensor data **132** of the vehicle cabin **206** including the occupants **202** and/or objects **214** in the vehicle cabin **206**. The vehicle sensor(s) **121** may capture the first sensor data **132** in any suitable format including images, video, and/or point cloud. As an example and as shown in FIG. 2B, the first sensor data **132** may be from the perspective of the vehicle sensor **121** and in an image format, in which a portion of the occupant **202** is visible and a portion of the occupant **202** is occluded by the laptop **204**. The first sensor data **132** may then be stored in the data store **115**.

[0042] The vehicle occupant safety determination system **100** may analyze the first sensor data **132** to determine the number of occupants **202** in the vehicle **102**, identify the occupant(s) **202** in the vehicle **102**, and/or determine whether the occupant(s) **202** are positioned safely such as sitting down properly and/or having their seatbelts **208** securely fastened. The vehicle occupant safety determination system **100** may utilize any suitable image processing, machine learning, and/or artificial intelligence methods or processes to make the determinations and/or identifications. The vehicle occupant safety determination system **100** may determine that a portion of one or more of the occupants **202** is occluded such that the vehicle occupant safety determination system **100** cannot make the determination of whether the occupant(s) **202** are positioned safely. In such a case, the vehicle occupant safety determination system **100** may search for a mobile device **204** to provide second sensor data **134** that may provide the portion of the occupant(s) **202** that is occluded in the first sensor data **132** such that the vehicle occupant safety determination system **100** may utilize the second sensor data **134** to determine whether the occupant(s) **202** are positioned safely.

[0043] The mobile devices **204** may include any device that is not a part of the vehicle **102** and/or is not permanently affixed to the vehicle **102** that is capable of capturing second sensor data **134**. As an example, the mobile device **204** may be a personal device such as a mobile phone, a computer, a laptop, a tablet, a camera, a display unit, and/or a video player. The mobile device **204** may be positioned such that the mobile device **204** may be capable of capturing second sensor data **134** that relates to the occupants **202** and/or other objects **214** in the vehicle cabin **206**. As such and

as an example, an occupant **202** may be holding the mobile device **204** or resting the mobile device **204** on their legs. Alternatively and as an example, the mobile device **204** may be affixed to a surface inside the vehicle cabin **206** such as on the back of a vehicle seat. In such an example, the mobile device **204** may be affixed to the vehicle seat in a temporary manner such that the mobile device **204** may be easily removed.

[0044] The mobile device(s) **204** may be capable of communicating, receiving, transmitting, and/or broadcasting data. The vehicle occupant safety determination system **100** may send a signal to the mobile device **204** requesting the second sensor data **134**. As an example, the vehicle occupant safety determination system **100** may send a signal to the mobile device **204** requesting the second sensor data **134**. In response, the mobile device **204** may capture and/or transmit the second sensor data **134** which may include information of the occupant(s) **202**, the vehicle cabin **206**, and/or objects **214** in the vehicle **102**. As another example, the vehicle occupant safety determination system **100** may send a signal to the mobile device **204** requesting to take control of the camera of the mobile device **204**. In such an example, in response to the mobile device **204** granting access to the vehicle occupant safety determination system **100**, the vehicle occupant safety determination system **100** may assume control of the camera of the mobile device **204**. Subsequently, the vehicle occupant safety determination system **100** may capture the second sensor data **134** in the form of images and/or video. As an example and as shown in FIG. 2C, the second sensor data **134** may be from the perspective of the sensor in the mobile device **204** and in an image format, in which the portion of the occupant **202** that is occluded by the laptop **204** is visible.

[0045] The vehicle occupant safety determination system **100** may analyze the second sensor data **134** to determine the number of occupants **202** in the vehicle **102**, identify the occupant(s) **202** in the vehicle **102**, and/or determine whether the occupant(s) **202** are positioned safely such as sitting down properly and/or having their seatbelts **208** securely fastened. The vehicle occupant safety determination system **100** may utilize any suitable image processing, machine learning, and/or artificial intelligence methods or processes to make the determinations and/or identifications.

[0046] With reference to FIG. 3, a more detailed block diagram of the vehicle occupant safety determination system **100** is shown. The vehicle occupant safety determination system **100** may include a processor(s) **110**. Accordingly, the processor(s) **110** may be a part of the vehicle occupant safety determination system **100**, or the vehicle occupant safety determination system **100** may access the processor(s) **110** through a data bus or another communication pathway. In one or more embodiments, the processor(s) **110** is an application-specific integrated circuit that may be configured to implement functions associated with a control module **320**. More generally, in one or more aspects, the processor(s) **110** is an electronic processor, such as a microprocessor that can perform various functions as described herein when loading the control module **320** and executing encoded functions associated therewith.

[0047] The vehicle occupant safety determination system **100** may include a memory **310** that stores the control module **320**. The memory **310** may be a random-access memory (RAM), read-only memory (ROM), a hard disk drive, a flash memory, or other suitable memory for storing the control module **320**. The control module **320** includes, for example, computer-readable instructions that, when executed by the processor(s) **110**, cause the processor(s) **110** to perform the various functions disclosed herein. While, in one or more embodiments, the control module **320** is a set of instructions embodied in the memory **310**, in further aspects, the control module **320** includes hardware, such as processing components (e.g., controllers), circuits, etc. for independently performing one or more of the noted functions.

[0048] The vehicle occupant safety determination system **100** may include a data store(s) **115** for storing one or more types of data. Accordingly, the data store(s) **115** may be a part of the vehicle occupant safety determination system **100**, or the vehicle occupant safety determination system **100** may access the data store(s) **115** through a data bus or another communication pathway. The data store(s) **115** is, in one embodiment, an electronically based data structure for storing information. In

at least one approach, the data store **115** is a database that is stored in the memory **310** or another suitable medium, and that is configured with routines that can be executed by the processor(s) **110** for analyzing stored data, providing stored data, organizing stored data, and so on. In either case, in one embodiment, the data store **115** stores data used by the control module **320** in executing various functions. In one embodiment, the data store **115** may be able to store sensor data **130** and/or other information that is used by the control module **320**.

[0049] The data store(s) **115** may include volatile and/or non-volatile memory. Examples of suitable data stores **115** include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. The data store(s) **115** may be a component of the processor(s) **110**, or the data store(s) **115** may be operatively connected to the processor(s) **110** for use thereby. The term “operatively connected” or “in communication with” as used throughout this description, can include direct or indirect connections, including connections without direct physical contact.

[0050] In one or more arrangements, the data store(s) **115** can include first sensor data **132**. The first sensor data **132** can originate from the sensor system **120** of the vehicle **102**. The first sensor data **132** can include data from vehicle sensors **121** such as radar sensors **123**, LiDAR sensors **124**, thermal sensors **125**, cameras **126**, and/or any other suitable sensors in the vehicle **102** that are capable of capturing inside the vehicle cabin **206**. The first sensor data **132** may include images, videos, and/or point clouds of the vehicle cabin **206**, occupants **202** in the vehicle cabin **206**, and/or objects **214** in the vehicle cabin **206**.

[0051] In one or more arrangements, the data store(s) **115** can include second sensor data **134**. The second sensor data **134** can originate from one or more mobile devices **204** located in the vehicle **102**. Similar to the first sensor data **132**, the second sensor data **134** may include images, videos, and/or point clouds of the vehicle cabin **206**, occupants **202** in the vehicle cabin **206**, and/or objects **214** in the vehicle cabin **206**.

[0052] In one embodiment, the control module **320** may include instructions that, when executed by the processor(s) **110**, cause the processor(s) **110** to request, in response to a portion of an occupant **202** being occluded in the first sensor data **132** captured from a vehicle sensor **121** in a vehicle cabin **206**, a second sensor data **134** from a mobile device **204** capable of capturing second sensor data **134** in the vehicle cabin **206**. In one or more arrangements, the control module **320** may receive first sensor data **132** from the vehicle sensor(s) **121**. As previously mentioned, the vehicle sensor(s) **121** may be one or more of a radar sensor **123**, a camera **126**, a LiDAR sensor **124**, a thermal sensor **125**, and/or any sensor suitable and capable of capturing sensor data **130** relating to the vehicle cabin **206**. The control module **320** may identify the occupant(s) **202** in the vehicle **102** based on the first sensor data **132**. The control module **320** may then determine whether there is a safety issue relating to one or more of the occupants **202**. A safety issue may include fastening of the seatbelt **208**, positioning of the seatbelt **208**, and/or positioning of the occupant **202**. In other words, the control module **320** may determine whether the occupant **202** has their seatbelt **208** securely fastened, whether the seatbelt **208** is properly positioned across the occupant's **202** body, and/or whether the occupant **202** is seated in a proper and safe manner based on the first sensor data **132**. As previously mentioned, the control module **320** may utilize any suitable image processing, machine learning, and/or artificial intelligence method to determine whether there is a safety issue relating to one or more of the occupants **202** of the vehicle **102**.

[0053] In one or more cases, an object **214** may be between the occupant **202** and the vehicle sensor(s) **121**. As such, the object **214** may be blocking the vehicle sensor's **121** view of the occupant(s) **202**. As an example, the occupant **202** may have a mobile device **204** such as a laptop blocking the view of the vehicle sensor(s) **121**, thus a portion of the occupant **202** may be occluded in the first sensor data **132**. In such a case, the control module **320** may be unable to determine

whether there is a safety issue and may request second sensor data **134** from the mobile device **204**. Alternatively and additionally, the control module **320** may request second sensor data **134** from another mobile device other than the mobile device **204** blocking the view of the vehicle sensor(s) **121**. In other words, the vehicle sensors' **121** view of the occupant(s) **202** may be occluded by an object **214**. In one case, the object **214** may be a mobile device **204** capable of capturing sensor data **130**. In another case, the object **214** may not be capable of capturing sensor data **130**. The control module **320** may detect one or more mobile devices **204** in the vehicle **102** and may then request second sensor data **134** from the mobile devices **204**.

[0054] In one or more arrangements, the control module **320** may detect multiple mobile devices **204** in the vehicle **102** and may then select one mobile device **204** from the multiple mobile devices **204**. The control module **320** may detect the mobile device **204** from the multiple mobile devices **204** in the vehicle cabin **206** using any suitable method. The mobile device **204** may be a mobile phone, a computer such as a laptop or a tablet, a display unit such as a video player, and/or a camera. The control module **320** may request and receive second sensor data **134** from the selected mobile device **204** and may determine whether the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134**. The control module **320** may utilize any suitable image processing, machine learning, and/or artificial intelligence techniques to determine whether the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134**. In a case where the control module **320** determines that the occluded portion is not visible in the second sensor data **134**, the control module **320** may select another mobile device **204** from the multiple mobile devices **204**. The control module **320** may then request and receive second sensor data **134** from the newly selected mobile device **204** and may determine whether the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134**. In a case where the control module **320** determines that the occluded portion is not visible in the second sensor data **134**, the control module **320** may continue to select other mobile devices **204** from the multiple devices **204**. The control module **320** may continue to request and receive second sensor data **134** from the selected mobile devices **204** until the control module **320** determines that the occluded portion is visible in the second sensor data **134**.

[0055] In one or more arrangements, the control module **320** may detect a mobile device **204** in the first sensor data **132**. The control module **320** may then determine the location of the mobile device **204** within the vehicle cabin **206** based on the location of the mobile device **204** in the first sensor data or using any other suitable methods. The control module **320** may then communicate with and establish a connection with the mobile device **204**. The control module **320** may request permission to connect to the mobile device **204** and may further request and receive second sensor data **134** from the mobile device **204**. In one case, the control module **320** may request and receive the second sensor data **134** from the mobile device **204**. In another case, the control module **320** may request and assume control of the mobile device **204**. The control module **320** may then capture the second sensor data **134** using the sensor(s) in the mobile device **204**.

[0056] In one embodiment, the control module **320** may include instructions that, when executed by the processor(s) **110**, cause the processor(s) **110** to determine, in response to the portion of the occupant **202** that is occluded in the first sensor data **132** being visible in the second sensor data **134**, whether there is a safety issue. As previously mentioned, the control module **320** may determine whether the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134** using any suitable method including image processing, machine learning, and/or artificial intelligence methods. The control module **320** may determine the seat that the occupant **202** with an occluded portion is sitting in in the vehicle **102** using information from sensors including other vehicle sensors such as seat sensors or weight sensors. The control module **320** may then determine whether the field of view of the second sensor data **134** includes a view of the determined seat. As such, the control module **320** may determine

whether the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134**.

[0057] Upon determining that the portion of the occupant **202** that is occluded in the first sensor data **132** is visible in the second sensor data **134**, the control module **320** may determine whether there is a safety issue relating to the occupant **202**. The control module **320** may utilize any suitable methods to determine whether the occupant **202** is positioned properly in their seat, e.g., sitting upright, whether the occupant **202** has their seatbelt **208** fastened, and/or whether the seatbelt **208** is positioned properly, e.g., a portion of the seatbelt **208** across the occupant's **202** pelvis and another portion of the seatbelt **208** across the occupant's **202** torso.

[0058] In a case where these conditions are met, the control module **320** may determine that there is no safety issue and the control module **320** may generate a notification indicating that there is no safety issue and the occupant **202** is securely fastened. In a case where one or more of these conditions are not met, the control module **320** may also generate a notification, indicating that there is a safety issue and may further indicate the safety issue, e.g., the occupant's **202** seatbelt **208** is not securely fastened, the straps of the seatbelt **208** are not properly positioned, and/or the occupant **202** is not sitting properly. The control module **320** may output the notification in any suitable manner including an audio output or a visual output such as an image, a video or a text.

[0059] Alternatively and/or additionally, in a case where these conditions are met, the control module **320** may determine that there is no safety issue, and the control module **320** may implement a vehicle action in response to no safety issue and the occupant **202** being securely fastened. In a case where one or more of these conditions are not met, the control module **320** may also implement a vehicle action in response to a safety issue such as the occupant's **202** seatbelt **208** not being securely fastened, the straps of the seatbelt **208** not being properly positioned, and/or the occupant **202** not sitting properly. The vehicle action may include starting the vehicle **102**, stopping the vehicle **102**, decelerating, and/or accelerating. As an example, the vehicle **102** may be stationary, and the control module **320** may determine that there is no safety issue before starting the vehicle **102** and/or activating the autonomous driving module(s) **160**. As another example, in response to determining that there is a safety issue and the vehicle **102** is in motion, the control module **320** may decelerate and/or stop the vehicle **102**. As another example, in response to determining that there is no safety issue and the vehicle **102** is in motion, the control module **320** may cause the vehicle **102** to accelerate or make a turn.

[0060] FIG. **4** illustrates a method **400** for determining a safety issue relating to an occupant **202** of a vehicle **102**. The method **400** will be described from the viewpoint of the vehicle occupant safety determination system **100** of FIGS. **1-3**. However, the method **400** may be adapted to be executed in any one of several different situations and not necessarily by the vehicle occupant safety determination system **100** of FIGS. **1-3**.

[0061] At step **410**, the vehicle occupant safety determination system **100** may request, in response to a portion of an occupant **202** being occluded in a first sensor data **132** captured from a vehicle sensor **121** in a vehicle cabin **206**, a second sensor data **134** from a mobile device **204** capable of capturing second sensor data **134** in the vehicle cabin **206**. As previously mentioned, the vehicle occupant safety determination system **100** may receive first sensor data **132** from one or more vehicle sensors **121**. The vehicle sensors **121** may be one or more of a radar sensor **123**, a camera **126**, a LiDAR sensor **124**, and/or a thermal sensor **125**.

[0062] The vehicle occupant safety determination system **100** may determine that portions of one or more occupants **202** in the vehicle **102** are occluded in the first sensor data **132**. The vehicle occupant safety determination system **100** may determine which seats in the vehicle **102** have occupants **202** with occluded portions using, as an example, one or more sensors or the first sensor data **132**. The vehicle occupant safety determination system **100** may then determine which mobile devices **204** are proximate to those seats and communicate with the mobile device(s) **204**, requesting second sensor data **134** from the mobile device **204** and/or requesting to take control of

the mobile device **204** so as to capture second sensor data **134**. In response to the request, the mobile device(s) **204** may transmit and the vehicle occupant safety determination system **100** may receive second sensor data **134**. The vehicle occupant safety determination system **100** may communicate and receive second sensor data **134** from one or more mobile devices **204**. The vehicle occupant safety determination system **100** may determine whether the portion of the occupant(s) **202** that is occluded in the first sensor data **132** is visible in any of the second sensor data **134** using any suitable image processing, machine learning, and/or artificial intelligence techniques.

[0063] At step **420**, the vehicle occupant safety determination system **100** may determine, in response to the portion of the occupant **202** that is occluded in the first sensor data **132** being visible in the second sensor data **134**, whether there is a safety issue. Upon receiving second sensor data **134** in which the occluded portion in the first sensor data **132** is visible, the vehicle occupant safety determination system **100** determines whether there is a safety issue. As previously mentioned, a safety issue may be related to the fastening of a seatbelt **208**, the positioning of the seatbelt **208** on the occupant **202**, and/or the positioning of the occupant **202**. As an example, the vehicle occupant safety determination system **100** may compare the images of the seatbelt **208** and/or occupant **202** to an image database to determine whether the seatbelt **208** is securely fastened, the seatbelt **208** is properly positioned, and/or the occupant **202** is properly positioned. The vehicle occupant safety determination system **100** may generate a notification and/or implement a vehicle action in response to the determination that there is a safety issue. Alternatively and/or additionally, the vehicle occupant safety determination system **100** may generate a notification and/or implement a vehicle action in response to the determination that there is no safety issue. The notification may be audio, visual, a vibration, or any other suitable notification method. The vehicle action may include starting the vehicle **102**, stopping the vehicle **102**, activating and/or deactivating a vehicle system such as autonomous driving modules, entertainment systems, accelerating, decelerating, and/or steering the vehicle **102**.

[0064] It will be appreciated that arrangements described herein can provide numerous benefits, including one or more of the benefits mentioned herein. For example, arrangements described herein can result in keeping and ensuring the safety of occupants of a vehicle.

[0065] Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. **1-4**, but the embodiments are not limited to the illustrated structure or application.

[0066] The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowcharts or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

[0067] The systems, components and/or processes described above can be realized in hardware or a combination of hardware and software and can be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. Any kind of processing system or another apparatus adapted for

carrying out the methods described herein is suited. A typical combination of hardware and software can be a processing system with computer-usable program code that, when being loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components and/or processes also can be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes described herein. These elements also can be embedded in an application product which comprises all the features enabling the implementation of the methods described herein and which when loaded in a processing system, is able to carry out these methods.

[0068] Furthermore, arrangements described herein may take the form of a computer program product embodied in one or more computer-readable media having computer-readable program code embodied, e.g., stored, thereon. Any combination of one or more computer-readable media may be utilized. The computer-readable medium may be a computer-readable signal medium or a computer-readable storage medium. The phrase “computer-readable storage medium” means a non-transitory storage medium. A computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer-readable storage medium would include the following: a portable computer diskette, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0069] Generally, modules, as used herein, include routines, programs, objects, components, data structures, and so on that perform particular tasks or implement particular data types. In further aspects, a memory generally stores the noted modules. The memory associated with a module may be a buffer or cache embedded within a processor, a RAM, a ROM, a flash memory, or another suitable electronic storage medium. In still further aspects, a module as envisioned by the present disclosure is implemented as an application-specific integrated circuit (ASIC), a hardware component of a system on a chip (SoC), as a programmable logic array (PLA), or as another suitable hardware component that is embedded with a defined configuration set (e.g., instructions) for performing the disclosed functions.

[0070] Program code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber, cable, RF, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present arrangements may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java™, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on a user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer, or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0071] The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . .” as used herein

refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC or ABC).

[0072] Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A system, the system comprising: a processor; and a memory storing machine-readable instructions that, when executed by the processor, cause the processor to: request, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin; and determine, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.
2. The system of claim 1, wherein the vehicle sensor includes one or more of: a radar sensor; a camera; a LiDAR sensor; or a thermal sensor.
3. The system of claim 1, wherein the safety issue includes one or more of: fastening of a seatbelt; positioning of a seatbelt; or positioning of the occupant.
4. The system of claim 1, wherein the machine-readable instructions further include instructions that when executed by the processor cause the processor to: detect the mobile device from a plurality of mobile devices in the vehicle cabin.
5. The system of claim 1, wherein the machine-readable instructions further include instructions that when executed by the processor cause the processor to: generate, in response to a safety issue, a notification.
6. The system of claim 1, wherein the machine-readable instructions further include instructions that when executed by the processor cause the processor to: implement, in response to a safety issue, a vehicle action.
7. The system of claim 1, wherein the mobile device includes one or more of: a mobile phone; a computer; a tablet; a display unit; or a camera.
8. A method, the method comprising: requesting, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a mobile device capable of capturing second sensor data in the vehicle cabin; and determining, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.
9. The method of claim 8, wherein the vehicle sensor includes one or more of: a radar sensor; a camera; a LiDAR sensor; or a thermal sensor.
10. The method of claim 8, wherein the safety issue includes one or more of: fastening of a seatbelt; positioning of a seatbelt; or positioning of the occupant.
11. The method of claim 8, further comprising: detecting the mobile device from a plurality of mobile devices in the vehicle cabin.
12. The method of claim 8, further comprising: generating, in response to a safety issue, a notification.
13. The method of claim 8, further comprising: implementing, in response to a safety issue, a vehicle action.
14. The method of claim 8, wherein the mobile device includes one or more of: a mobile phone; a computer; a tablet; a display unit; or a camera.
15. A non-transitory computer-readable medium including instructions that when executed by a processor cause the processor to: request, in response to a portion of an occupant being occluded in a first sensor data captured from a vehicle sensor in a vehicle cabin, a second sensor data from a

mobile device capable of capturing second sensor data in the vehicle cabin; and determine, in response to the portion of the occupant that is occluded in the first sensor data being visible in the second sensor data, whether there is a safety issue.

16. The non-transitory computer-readable medium of claim 15, wherein the vehicle sensor includes one or more of: a radar sensor; a camera; a LiDAR sensor; or a thermal sensor.

17. The non-transitory computer-readable medium of claim 15, wherein the safety issue includes one or more of: fastening of a seatbelt; positioning of a seatbelt; or positioning of the occupant.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions further include instructions that when executed by the processor cause the processor to detect the mobile device from a plurality of mobile devices in the vehicle cabin.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions further include instructions that when executed by the processor cause the processor to generate, in response to a safety issue, a notification.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions further include instructions that when executed by the processor cause the processor to implement, in response to a safety issue, a vehicle action.
